CAMPUSFLEX

"A Campus which you have to survive."

Project Team

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Section D

SDA Project Proposal - Spring 2026

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February, 2026

1. Project Overview

CampusFlex is a 2D Role-Playing Game (RPG) developed in Java that presents a fictionalized version of student life at FAST-NU Islamabad. The game combines an open-world campus simulation with a story-driven survival mode to reflect both the academic routine and emotional challenges faced by FAST students.

In the first phase, called *Normal Mode*, players live through a typical semester. They attend classes, study in the library, participate in events, socialize, visit the mosque, manage sleep, and interact with their mobile phones. Each activity affects core player stats such as GPA, Energy, Stress, Sleep, and Karma. Every decision comes with trade-offs, encouraging players to balance academic performance with personal well-being.

During the "dead week", a failure in the university's AI lab traps the campus in an infinite loop, turning students and staff into mindless zombies; hence starting the zombie mode. The player must navigate familiar campus locations to reach the server room and deploy a system patch.

Combat is academic in nature. Players may choose to "Debug" enemies by solving logic and programming challenges or "Terminate" them through brute force. Actions taken during Normal Mode directly affect Zombie Mode. Helping classmates, skipping lectures, or cheating in quizzes all influence how the player is treated later. The game ends with the player saving the campus and passing the semester without sitting final exams, reinforcing the core philosophy: *what goes around comes around* at Fast.

2. Motivation

FAST-NU is known for its demanding academic environment, which often overwhelms new students and leads to stress, burnout, and gaps in fundamental concepts such as Programming Fundamentals. Traditional teaching methods rarely address life balance or emotional fatigue.

CampusFlex uses gamification to bridge this gap. By turning daily student routines and academic concepts into interactive gameplay, the project offers both a psychological escape and a learning experience. Players practice logical thinking while also learning the importance of managing time, stress, and energy.

Rather than building generic systems, CampusFlex targets a specific audience: FAST students. Familiar locations, inside jokes, and recognizable behaviors make the game relatable and engaging. This localized approach also provides a practical opportunity to test software scalability, user behavior, and feedback, allowing the project to grow beyond a semester assignment into a deployable product.

3. Objectives

The main objectives of CampusFlex are:

- To design and implement a scalable RPG architecture in Java using clean coding practices and standard Software Design Patterns.
- To simulate student life through meaningful gameplay mechanics such as GPA, Energy, Stress, Sleep, and Karma.

- To demonstrate the “actions have consequences” principle by connecting Normal Mode behavior directly with Zombie Mode difficulty.
- To develop a deployable prototype showcasing exploration, stat management, academic combat, and progression.
- To reinforce foundational programming and logical thinking through interactive challenges.
- To provide students with a relatable platform that promotes healthier academic habits and life balance.

4. Expected Outcomes

By the completion of this project, the team aims to deliver:

- A playable 2D RPG prototype featuring both Normal Mode and Zombie Mode.
- A standalone executable (.jar) file that runs on any Java-enabled desktop system.
- A complete system design including UML diagrams and gameplay flow structures.
- A working stat system where player choices meaningfully impact progression.
- At least one fully explorable campus zone with NPC interactions and academic combat.
- A stable gameplay experience that successfully blends entertainment with educational reinforcement.

5. Project Scope

- Open-world campus simulation with core student activities.
- Player stat management (GPA, Energy, Stress, Sleep, Karma).
- Academic combat with logic and programming challenges.
- Consequence-based gameplay linking both modes.
- FAST-themed inside references and humor.
- Java-based game architecture.

6. Tools and Technologies

CampusFlex is developed using the following tools and technologies:

- **Frontend:** Java and JavaFX are used to build the graphical user interface, handle animations, and manage player interactions within the game.
- **Backend:** Oracle is used for storing player data, stats, progress, and game states, enabling persistence across sessions.
- **Version Control:** GitHub is used for version control, allowing collaborative development, code tracking, and safe integration of features.
- **Project Management:** ClickUp is used to manage tasks, assign responsibilities, track progress, and follow an organized development workflow.